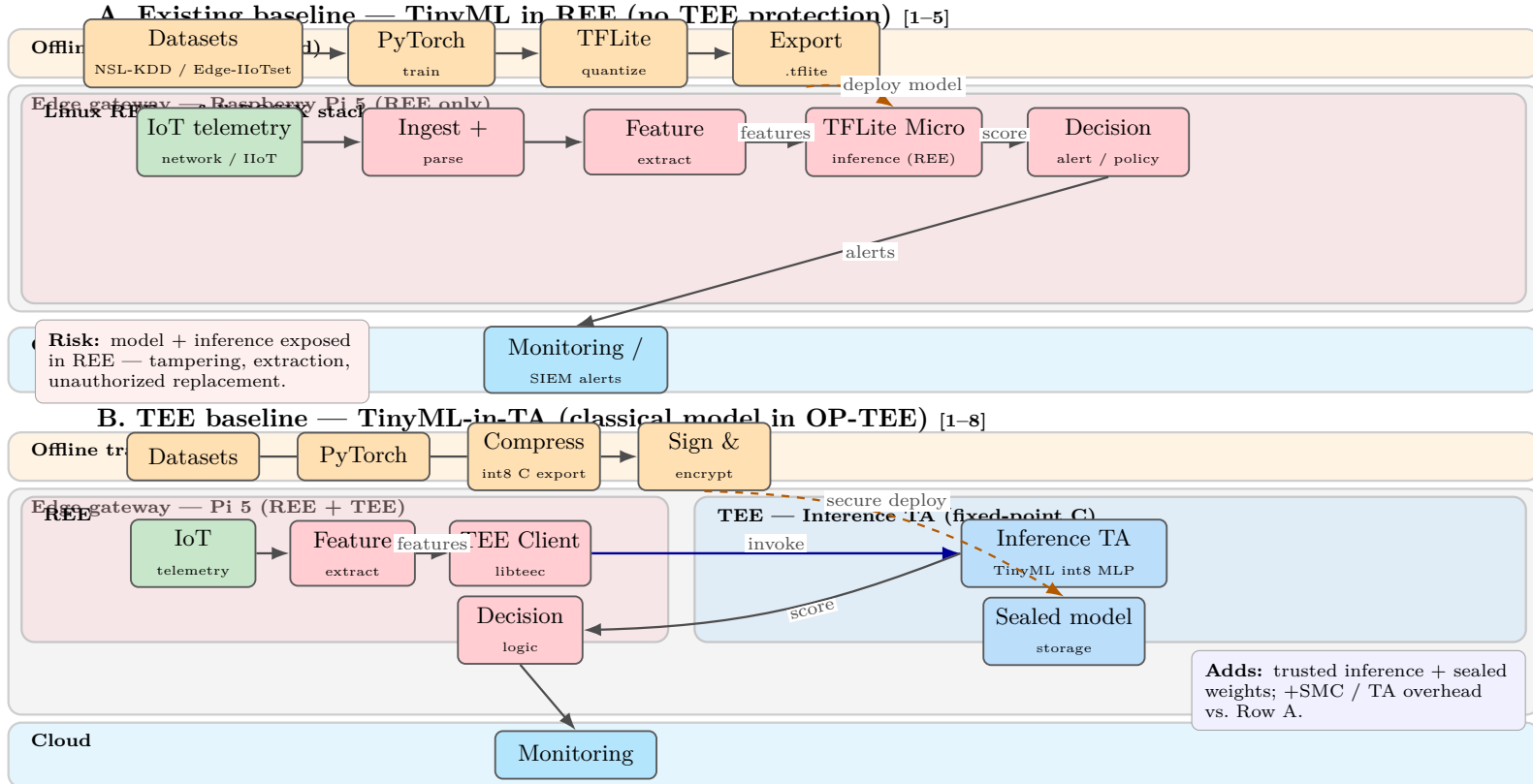
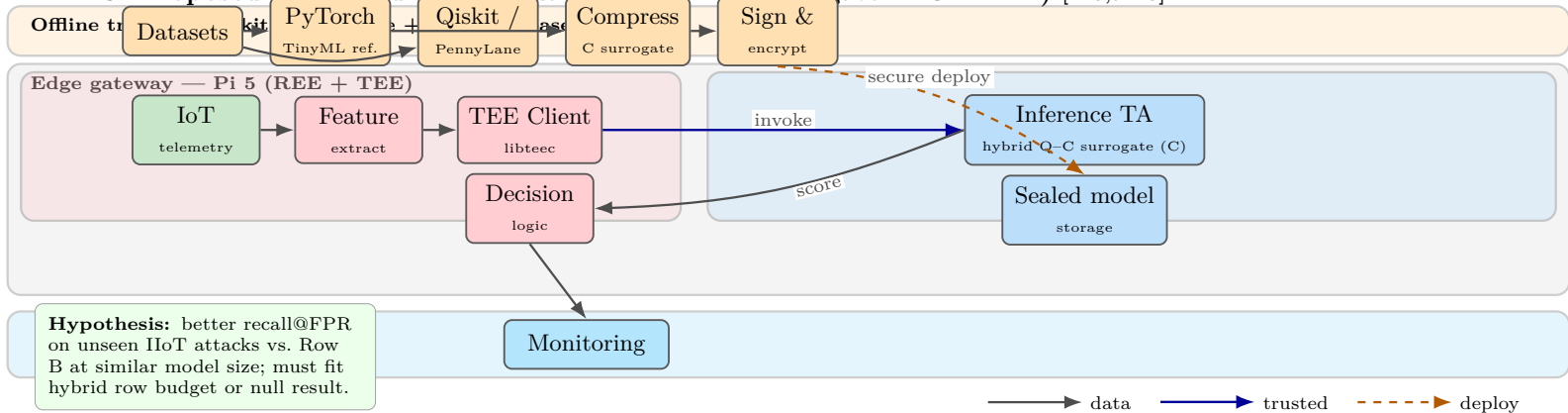


Baseline vs. Proposed Edge Pipelines

Same gateway workload (5–20 scoring events/s); comparative flows on Raspberry Pi 5 — budgets aligned with [tee_qai_edge_architecture.pdf](#) Page 3



C. Proposed — Hybrid-in-TA (quantum-inspired surrogate in OP-TEE) [1–5,9–13]



Comparative budget — Rows A / B / C (engineering targets)

Shared scenario: NSL-KDD ($N=41$) [4] or Edge-IIoTset ($N\approx63$) [5]; 5–20 scoring events/s; int8 models $\sim 5\text{--}30\text{ K}$ parameters. Cost metrics follow PICO TinyML benchmark [3] and MLPerf inference/power guidance [13]. Values are engineering targets pending Pi 5 measurement.

End-to-end latency (ms per scoring event, session kept open)

Pipeline	Typical	Max	TEE?	Notes
A. TinyML in REE (TFLite)	2	8	No	Fastest; no model protection
B. TinyML-in-TA (same weights)	8	15	Yes	+world switch / SMC overhead
C. Hybrid-in-TA (surrogate)	20	35	Yes	Larger compute; qubit depth capped at train time

CPU duty cycle (one Cortex-A76 core)

Pipeline	ms/event	@ 5 events/s	@ 20 events/s
A. TinyML in REE	~ 2	$\sim 1\%$	$\sim 4\%$
B. TinyML-in-TA	~ 8	$\sim 4\%$	$\sim 16\%$
C. Hybrid-in-TA	~ 20	$\sim 10\%$	$\sim 40\%$

Duty cycle $\approx (\text{ms/event} \times \text{rate}) / 1000$. Row C at 20/s uses $\sim 40\%$ of one core — acceptable if gateway is not line-rate IPS.

Memory footprint (production C path)

Resource	Row A	Rows B & C	Comment
REE daemon RSS	4–8 MB	4–12 MB	Row B/C add libteec client
Model storage location	REE filesystem	TEE sealed store	Row A vulnerable to tampering
Model size (int8)	10–80 KB	10–80 KB (B); 20–200 KB (C)	C may be slightly larger; cap at train time
TA_DATA_SIZE	n/a	256–512 KB	Required for B/C; OP-TEE default 32 KB too small
OP-TEE SHM + TA pool	n/a	$\sim 4 + 8\text{--}16$ MB	One-time platform cost for TEE rows

Security & evaluation role

	A (existing)	B (TEE baseline)	C (proposed)
Model integrity	Not protected	Sealed + attestation	Same as B
Detection engine	Classical TinyML	Classical in TA	Hybrid Q–C surrogate
Benchmark role	Speed / accuracy reference	Trust overhead reference	QML value test vs. B
Accept criterion	—	Latency $\leq 4\times$ Row A [3]	Recall@FPR beats B [5][9] within C budget

Decision rule (head-to-head)

- **Row A** establishes unconstrained classical performance (upper bound on speed, lower bound on trust).
- **Row B** quantifies the cost of TEE protection for the *same* TinyML model.
- **Row C** is justified only if hybrid-in-TA improves recall@FPR (Edge-IIoTset stealthy/unseen attacks) versus Row B **without exceeding** ~ 35 ms max latency and $\sim 40\%$ core duty at 20 events/s — otherwise publish null result.

Prototype order: Row A harness (REE TFLite) $\rightarrow$ Row B (passthrough TA + TinyML TA) $\rightarrow$ Row C (hybrid surrogate TA, Year 2).

References and cost-metric standards

Pipeline mapping (Rows A / B / C)

Row A (TinyML in REE): TFLite Micro runtime [1], TinyML practice [2], PICO cost metrics [3], datasets [4][5].
Row B (+ TEE): OP-TEE TA model [6][7][8], same TinyML export path as A; adds secure-world inference overhead.
Row C (+ hybrid QML): Chowdhury *et al.* [9], PennyLane [10], D’Amore *et al.* [11], embedded QML feasibility [12]; builds on B with offline QML training and C surrogate in TA.

Standard cost and evaluation metrics (this study)

Metric	Definition / target	Standard / reference
Inference latency	Wall-clock ms per scoring event (session open)	PICO-tinyML-benchmark [3]; <code>clock_gettime</code>
End-to-end latency	Ingest $\rightarrow$ alert; target ≤ 100 ms	Edge IDS SLA practice [5]
CPU duty cycle	$(\text{ms/event} \times \text{rate}) / 1000$ on one core	PICO CPU utilization [3]
REE RSS memory	CA + buffers (MB)	<code>/proc/self/status</code> ; PICO memory efficiency [3]
TA heap / stack	TA_DATA_SIZE, TA_STACK_SIZE	OP-TEE TA properties [6][7]
Model size	Sealed int8 weights (KB)	TinyML compression [1][2]
Energy / inference	Joules or mJ per scored flow	MLPerf power methodology [13]
Detection quality	Recall@FPR (e.g. 1%), AUC, F1	Edge-IIoTset IDS evaluation [5]; family hold-out for unseen attacks
Trust overhead	B vs A latency ratio ($\leq 4\times$ target)	TEE SMC + TA invoke [6][8]

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